

Quan Do

Urbana, IL | quando2@illinois.edu | (669) 369-9547
[linkedin.com/in/quan-m-do/](https://www.linkedin.com/in/quan-m-do/)

EDUCATION

University of Illinois Urbana-Champaign | Champaign, IL Expected May 2028

Master of Science in Computer Science

Relevant coursework: Applied Machine Learning, Distributed Systems, Networked IoT Systems

University of Massachusetts Amherst | Amherst, MA

May 2026

Bachelor of Science in Computer Engineering

Mathematics minor | GPA: 3.84; Dean's List

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, SQL, Bash, Assembly, HTML/CSS

Embedded & perception: AUTOSAR, GPIO/UART, timer interrupts, Raspberry Pi 5, YOLOv8n, KeilC

Tools & frameworks: Git, GitHub, Jira, MATLAB, Simulink, Verilog, Node.js, Express, GraphQL

WORK EXPERIENCE

Computer Support Technician | UMass EdTech, Amherst, MA

May 2025 - May 2026

- Resolved hardware/software issues for 150+ users monthly through Jira, achieving a 96% resolution rate within agreed response times.
- Built Power BI dashboards and automated Qualtrics/Power Automate workflows, reducing reporting time by 40%.

Embedded Software Engineer Intern | FPT Software | Ho Chi Minh City, Vietnam

Jun 2024 - Aug 2024

- Developed AUTOSAR-compliant systems in C on the S32K144 EVB using KeilC; optimized LPUART interfaces at 9600/19200 baud, achieving 100% communication accuracy.
- Integrated the SysTick timer for interrupt-driven scheduling, reducing latency by 25% and supporting real-time performance in an automotive system.
- Led a 5-week project implementing GPIO, clock peripherals, and timers; achieved 30% faster development and delivered a scalable prototype ahead of schedule.

Software Engineer Intern | Infina (YC S21) | Ho Chi Minh City, Vietnam

May 2023 - Aug 2023

- Built a Node.js, TypeScript, GraphQL, and Express API backend handling 1,000+ requests/second; tripled order throughput with sub-50 ms responses for a platform serving 500,000+ investors.
- Implemented phone-number and unique savings-code validation, reducing user errors by 30% and achieving 100% data integrity for those fields.
- Improved code quality with TypeScript and TSDoc, supporting monthly releases by a 4-engineer team.

PROJECTS

RunRoute | Personal Project

August 2026 - Present

- Built a React/TypeScript route planner over 116,000+ OpenStreetMap nodes in Urbana-Champaign, supporting loop and point-to-point routes with interactive Leaflet maps, terrain preferences, and traffic-light avoidance.
- Implemented A* search in a Web Worker with quality gates enforcing route lengths 0-5% above target and at most 1% repeated geometry; added GPX export and regression tests for routing constraints and distance consistency.

Animal Detector on Road | Senior Design Project

Sep 2025 - May 2026

- Led the perception and detection team for a Raspberry Pi 5-based RC-car vision system that detected deer-sized animals and triggered driver alerts and braking.
- Collected and labeled onboard video; trained a YOLOv8n detector with data augmentation and deployed a quantized model with distance-based braking logic and emergency stops via GPIO/UART.

Typing Speed Test Web Application | Personal Project

Apr - Aug 2024

- Built an HTML, CSS, and JavaScript web app integrating the Advice Slip API for randomized prompts and a WPM calculator with 95-98% accuracy.
- Deployed on Netlify and maintained source code with GitHub for version control and updates.